

Swapnil Tripathi | PhD Candidate |

Power Electronics | UC Santa Cruz

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Objective

PhD researcher in power electronics with hands-on experience taking high-voltage converters from topology concept through simulation, multi-layer PCB design, and bench validation. Seeking a **Power Electronics Internship** to contribute experience in **high step-down converter design across 1000 V-to-48 V class buses, GaN/SiC/IGBT device characterization and gate driver design, and grid-tied inverter control (PR + PLL)** to scalable, efficient medium-voltage power conversion systems.

Education

Program	Institution	GPA	Year
PhD (Power Electronics)	University of California, Santa Cruz	4.0	2025 – Present
M.S. (Power Systems & Electronics)	IIT Madras, Chennai, Tamil Nadu	3.8	2022 – 2025
B.Tech (Electrical Engineering)	MITS, Gwalior, Madhya Pradesh	3.55	2018 – 2022

Areas of Expertise

- **High- & Medium-Voltage Converter Topologies:** Hybrid switched-capacitor (HSC) networks (Dickson, SDIH, series-capacitor buck), multilevel and cascaded inverters, buck/boost, design experience spanning **1000 V-class DC buses down to 48 V and lower rails**, with per-device stress management and soft-charging/ZVS techniques.
- **Switching Device & Magnetics Characterization:** GaN HEMTs, Si/SiC MOSFETs and IGBTs device selection by $R_{DS(on)}/C_{OSS}$ figure of merit, C_{OSS} loss budgeting, dead-time and ZVS transition tuning, gate-loop and switch-node ringing analysis; inductor and flying-capacitor sizing, derating, and loss allocation.
- **Hardware Design & Bring-Up:** Schematic capture and multi-layer PCB layout in Altium Designer (2 oz and 1 oz Cu stack-ups), component selection and BOM, DRC, board bring-up and fault debug.
- **Simulation & Modeling:** LTspice, PSpice, MATLAB/Simulink (converter-level circuit simulation), control-loop modeling, and loss/efficiency estimation.
- **Control Systems:** PR (proportional-resonant) current control, PLL grid synchronization, field-oriented control (FOC), V/f, SPWM/SVPWM; digital implementation on TI C2000 DSP and Xilinx FPGA.
- **Bench Test & Validation:** High-voltage bench testing under documented safety practice, efficiency curve measurement, switching-waveform capture and analysis, thermal profiling, and design-review documentation.

Key Projects

1. 800 V-to-28 V GaN HSC Converter — Space and Aerospace Power2026
- Designed and validated a **420 W, 800 V → 28 V** stacked-Dickson Hybrid Switched Capacitor converter merged with a multi-phase buck regulation stage at **250 kHz**, targeting **lunar and planetary surface power distribution** and **aerospace high-voltage DC tether systems**, where mass, efficiency, and radiation tolerance are the governing constraints.
 - Architecture distributes the bus across a multilevel switch stack so per-device stress stays a fixed fraction of the input, enabling **low-voltage (150 V) GaN HEMTs on an 800 V bus** — a natural derating that addresses single-event burnout margins for space-qualified devices.
 - Implemented **resonant charge redistribution (RCR)** to impose ZVS across all Dickson-stage switches, cutting CV^2f switching loss and measurably reducing switch-node overshoot and ringing; achieved **92.96 % peak power-stage**

and **90.66 % peak system efficiency**.

- Full hardware ownership: topology analysis, device and magnetics selection, schematic capture, **4-layer 2 oz Cu PCB layout in Altium**, assembly, and bench validation — **128 g total mass, 3.28 kW/kg** gravimetric power density, roughly $3\times$ that of comparable published high step-down converters.
- Bench work included flying-capacitor voltage verification across the stack, switch-node and inductor-current waveform capture, and efficiency sweeps at 400/600/800 V input with and without gate-driver loss accounted.

2. 48 V-to-1 V Switching-Bus Converter — Full-ZVS RCR 2026 – Present

- Proposed an **enhanced RCR clocking scheme** that pre-energizes the resonant inductor ahead of each commutation, achieving **complete ZVS** on the front-end 2:1 stage where prior art achieved only partial ZVS, no added hardware, purely a control and timing contribution.
- Derived the pre-energization current and interval (I_0 , Δt) from an LC energy-balance argument and verified the scheme on a **50 W, 1 MHz, 48 V $\rightarrow$ 1 V hardware prototype**.
- Built on a **6-layer 1 oz Cu Altium PCB** (symmetric dual-inductor hybrid Dickson stage, 16 switches, dense flying-capacitor bank on reverse side); handled gate driver and isolated level-shifter selection, layout for minimum commutation-loop inductance, and full bring-up.
- Measured efficiency improvement from **86.3 % to 88.6 %** at 20 A and **76.6 % to 85.4 %** at 5 A with RCR enabled; work carried out with **Abacus Power** (startup) on ultra-high-density point-of-load delivery for data-center power architectures.

3. Grid-Tied Inverter Control — PR + PLL (IISc) Feb – Jun 2025

- Implemented **proportional-resonant (PR) current control** with **PLL-based grid synchronization** on a single-phase H-bridge Nova converter for a **grid-tied energy storage** application.
- Developed the control loop in firmware, tuned it against the plant model, interfaced the converter to the grid, and closed the loop in hardware including oscilloscope-based waveform analysis, current/voltage measurement, and steady-state and transient validation.

4. 10 kVA Cascaded Multilevel Inverter — HV Drive (MS) 2022 – 2025

- Proposed and validated a novel **capacitor voltage balancing PWM** method for single-DC-source cascaded three-level inverters, published at **IEEE ECCE 2024** and **IEEE IECON 2024**.
- Built and commissioned a **10 kVA cascaded multilevel inverter** (>400 V DC bus) driving a **1.5 kW induction motor** on a hybrid **DSP + FPGA** platform (TMS320F28377S + Xilinx FPGA); extensive hands-on work with oscilloscopes, high-voltage differential and current probes, and power analyzers.
- Implemented and validated **indirect vector control (FOC)** and V/f control in both MATLAB/Simulink and hardware; debugged gate-drive timing, DC-bus ripple, and motor current waveforms on the bench.

5. Gate Driver, Sensing & Inverter Hardware 2023 – 2025

- Designed a **10 kVA Si-IGBT (IKW15N120T2) gate driver** using ACPL-333J with **active Miller clamp, desaturation detection**, and hardware overcurrent/overvoltage protection core building blocks for reliable high-voltage switching.
- Designed a **3-phase 10 kW inverter** (200–650 V AC motors) around an IXYS MIXA60W1200TED module and UCC21530 dual-output gate driver; owned schematic capture, layout, bring-up, and validation.
- **Team Raftar Formula Racing (IIT Madras / Daimler Trucks)**: Designed the **signal conditioning and sensing board** for an IGBT high-voltage inverter driving a 5 kW PMSM — ADC front end, fault detection, TMS320F28379D MCU, PWM output stage, and CAN/SPI interfaces.
- **Mentor, Team Avishkar Hyperloop (IIT Madras)**: Guided H-bridge converter design with embedded current sensing and gate drive for the levitation module, and a 3-phase inverter for the traction module.

Publications

- **Swapnil Tripathi**, et al., “A High Step-Down Hybrid Switched-Capacitor DC–DC Converter for High Voltage Space Applications,” *Power Electronics for Aerospace Applications (PEASA)*, 2026. **(Accepted & Presented)**
- **Swapnil Tripathi**, et al., “An Optimized Resonant Charge Recovery Clocking Scheme for 48 V-to-1 V Switching Bus Converters,” *IEEE Applied Power Electronics Conference & Exposition (APEC)*, 2027. **(Submitted)**

- **Swapnil Tripathi**, Uddhav Surve and Srirama Srinivas, "Capacitor Voltage Balancing Pulsewidth Modulation Method for a Single DC Source Driven Cascaded Three-level Inverter Drive," *IEEE Energy Conversion Congress & Exposition (ECCE)*, 2024. **(Published)**
- **Swapnil Tripathi**, Uddhav Surve and Srirama Srinivas, "Carrier-Modified Pulsewidth Modulation for a Single DC Power Source Driven Cascaded Three-level Inverter Drive," *Annual Conference of the IEEE Industrial Electronics Society (IECON)*, 2024. **(Published)**
- **Swapnil Tripathi**, Ashwaini Goswami, "Wireless Power Transfer," *International Journal of Scientific Research in Engineering and Management (IJSREM)*, 2022. **(Published)**

Technical Skills

- **Simulation:** LTspice, PSpice (converter circuit and switching-transient simulation); MATLAB/Simulink (topology, control-loop, and system-level modeling); LabVIEW.
- **PCB & Schematic:** Altium Designer (primary), KiCad, 4 and 6-layer stack-ups in 1 oz and 2 oz Cu, high- di/dt commutation-loop layout, creepage/clearance for high-voltage nodes, DRC, BOM generation.
- **Devices & Magnetics:** GaN HEMTs (EPC), Si and SiC MOSFETs, IGBT modules; isolated and non-isolated gate drivers (LM5114, UCC21530, ACPL-333J, ADuM5240); inductor and flying-capacitor selection, derating, and loss modeling.
- **Lab & Validation:** Oscilloscopes, high-voltage differential and current probes, power analyzers, network analyzers, signal generators, electronic loads efficiency curve measurement, thermal profiling, switching-waveform analysis, and hands-on high-voltage bench practice.
- **Embedded & Control:** TI C2000 DSP (TMS320F28377S, F28379D), Xilinx FPGA (Vivado); CAN, SPI, I2C; PR, PI, and FOC control loop implementation.
- **Programming:** C/embedded C, Python, MATLAB, Code Composer Studio, Arduino including scripting for measurement post-processing and test data analysis.

Experience

- Graduate Researcher — UC Santa Cruz** **2025 – Present**
 - High-voltage and high-density hybrid switched-capacitor converter research: two hardware prototypes (800 V-to-28 V, 420 W GaN; 48 V-to-1 V, 50 W) designed, fabricated, and validated end-to-end; two first-author conference papers resulting.
- Project Associate — IISc Bengaluru** **Feb – Jun 2025**
 - Grid-tied energy storage inverter: PR current controller, PLL grid synchronization, schematic and hardware design, bring-up, and validation on a 4-legged Nova converter.
- Powertrain Engineer — Team RFR, IIT Madras** **2023 – 2025**
 - Designed sensing and gate driver boards for a high-voltage IGBT inverter driving a 5 kW PMSM; collaborated cross-functionally with Daimler Trucks engineers on system integration, design review, and validation.
- Industrial Training — NSPL, Odisha** **Jan – Apr 2022**
 - Hands-on exposure to a 10 MW steam power plant: medium-voltage switchgear, power drives, protection systems, and turbine auxiliary maintenance — practical grounding in utility-scale equipment and high-voltage safety practice.
- Robotics Engineer — Asimov Robotics Club, MITS** **2018 – 2022**
 - Designed an autonomous obstacle-avoidance robot (ultrasonic sensing, Arduino UNO); developed real-time navigation and path-finding algorithms.

Course Work

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| PhD Courses | 2025 – Present |
| <ul style="list-style-type: none"> • Electric Motor Drives, Advanced Analog Circuits, CMOS RF Circuit Design, Advanced Topics in ECE. | |
| MS Courses | 2022 – 2025 |

- Power Converter Analysis and Design, Power Electronics and System Design, Digital Controller for Power Applications, Power Electronic Control of Electric Motors, Power Electronics for Electric Vehicles and Drives.

B.Tech Courses

2018 – 2022

- Networks and Circuit Theory, Electromagnetic Field Theory, Analog & Digital Electronic Circuits, Electrical Machines, Measurement and Instrumentation, Power Systems, Control Systems, Power Electronics, Signals and Systems, Microprocessors.

Certifications

- **Coursera (CU Boulder):** Introduction to Power Electronics; Converter Circuits. (2023–2024)
- **NPTEL:** Design of Power Electronic Converters, Power System Dynamics, Automatic Control, Renewable Energy Engineering (Solar, Wind & Biomass), Advances in UHV Transmission and Distribution, Non-conventional Energy Resources, Control and Monitoring. (2018–2022)

Scholarships & Awards

- **Ministry of Human Resource Development (MHRD) Scholarship, India** — Awarded for exceptional performance in GATE 2022 with a national rank of **650** in Electrical Engineering.
- **Silver Medal** — Inter Hostel Basketball Tournament, IIT Madras.
- **3rd Place** — Mobile Photography Competition (Independence Day theme), IIT Madras.